

Fixed-Point Grasping with RoboMaster EP

Experiment 2 Records and Result Analysis

MAIN SIMULATION
5 / 5 cycles

MEAN SIMULATION ERROR
4.358 mm

MAX. SIMULATION ERROR
5.604 mm

HARDWARE: TEAM-OBSERVED
5 / 5 transfers

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Contents

1	Experiment Requirements and Individual Scope	2
2	Principles of a Verifiable Grasping Experiment	3
3	Record Audit and Reproducibility	4
4	Simulation Results and Error Interpretation	5
5	Hardware Outcome and Simulation Differences	6
6	Problems, Solutions and Individual Reflection	7
7	Shared GitHub Repository and Commit Evidence	8
8	Sources, Reproduction and Evidence Limits	9

Reading the results. The millimetre values on the cover come from the main simulation record. The hardware outcome is the team's confirmation of five successful transfers with no observed abnormalities; it is not a measurement of physical placement precision.

This individual report focuses on experiment records and result analysis. Its content is based on the reviewed Experiment 2 report and the matching project archive. The QYY personal template supplies the visual design only. Body page numbering starts with this contents page; the cover is unnumbered.

1 Experiment Requirements and Individual Scope

1.1 Task and platform adaptation

The task is fixed-point grasping without visual localization: home, approach A, descend, grip, lift, transfer to B, release and return. The course handout requires a complete simulation launch, state publication, saved trajectories/results/errors, joint and reachability checks, and at least four successful transfers in five consecutive trials. Hardware work additionally requires supervised low-speed operation, safe failure handling and individual safety-operation checks [1].

Because of equipment availability, our group used the DJI RoboMaster EP arm and gripper instead of the specified mechArm 270. The simulation uses ROS 2 Humble, Gazebo Fortress and custom kinematics/control on a Jetson Linux system. The physical implementation uses the RoboMaster Python SDK. This is a platform adaptation, not an unchanged ROS 2 hardware backend.

1.2 My responsibility and the team's interfaces

My main responsibility is organizing simulation results and report material. I use the files supplied by the simulation members to summarize the five trials, calculate placement errors and explain the results. The hardware pair provides the physical-test outcome. My role does not include operating the real robot or claiming authorship of the simulation code.

Table 1: Shared division of work and my reporting scope

Member	Work area	Main responsibility
Ning Bo	Simulation	Environment and scene setup; model loading and launch.
Huang Junbiao	Simulation	Motion programming; grasp, transfer and repeated-cycle debugging.
Yongyue Qi	Simulation results	Log organization, numerical analysis and report material.
Xue Yutong	Hardware	Pick/release waypoints, approach/lift heights, gripper adjustment and retention checks; joint trials.
Bao Lintai	Hardware	SDK communication, task sequence, chassis turn/return and exceptions; joint trials.

The two hardware members share physical setup, parameter review and integrated five-cycle testing. Their technical focuses differ, but neither is assigned a permanent passive-helper role. Each student remains responsible for the analysis in their own individual report.

Scope of evidence. Simulation repeats A→B and verifies home after every cycle, with object replenishment between cycles. Hardware alternates A/B, pauses for human grasp confirmation and returns home after all five cycles. The shared-interface/parameter-only migration objective and each member's safety-operation check are not demonstrated by the supplied records.

2 Principles of a Verifiable Grasping Experiment

2.1 From motion to acceptance evidence

The simulation sequence is home, hover, descend, close, hold, lift, turn, lower, open and retreat. Placement verification is followed by heading restoration and arm homing. Only then is a successful cycle appended to the results. A box reaching B alone does not prove completion of the full cycle, including return.

The scene contains a 25 mm cube and fixed world points $A = (0.278, -0.001, 0.0725)$ m and $B = (0.237, 0.1453, 0.0725)$ m. Fresh bilateral contact feedback triggers a small closure preload. HOLD has a nominal 3 s duration, followed by verification conditions that must remain satisfied for at least 0.4 s before lifting. Transfer uses chassis rotation plus small position corrections; the arm is not treated as a six-axis industrial manipulator.

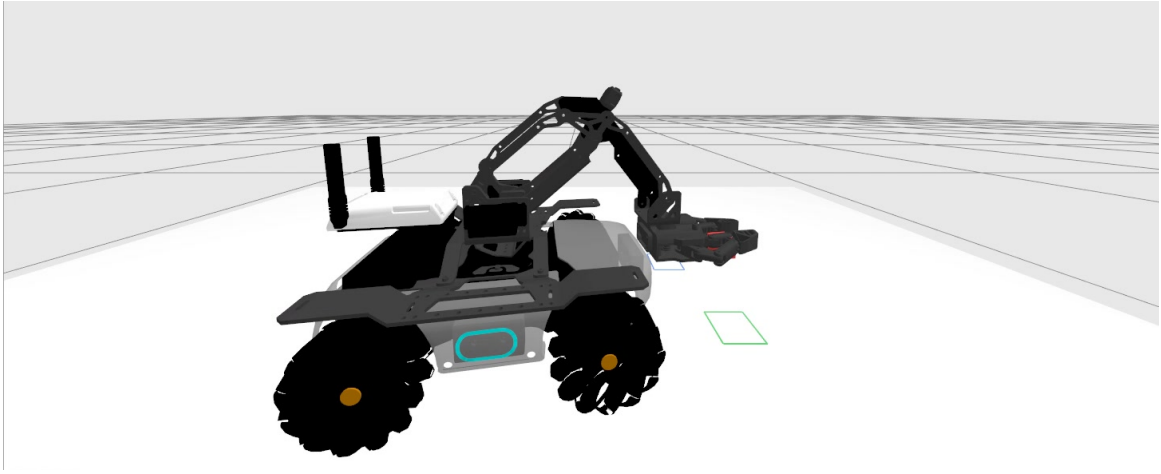


Figure 1: Scene frame from the team's Experiment 2 simulation recording at 34 s. This is a simulation view, not a photograph of the hardware or an error measurement.

2.2 Definitions and units

For recorded placement (x_i, y_i, z_i) and target (x_B, y_B, z_B) in metres, the horizontal placement error is

$$e_{xy,i} = 1000\sqrt{(x_i - x_B)^2 + (y_i - y_B)^2} \quad \text{mm}, \quad \bar{e} = \frac{1}{5} \sum_{i=1}^5 e_{xy,i}. \quad (1)$$

The controller's configured acceptance test requires $e_{xy} < 23$ mm, $|z_i - z_B| < 8$ mm and recent table contact. These are program thresholds, not millimetre tolerances specified by the course handout. The sample mean is not a guaranteed accuracy specification. Heading error is the absolute difference from the initial chassis heading at the home check.

Cycle duration uses a monotonic wall clock and includes execution and waiting under the actual computer load. The first cycle includes initial homing; later cycles start after refill. The first cycle's additional time should not be interpreted as degradation in grasping performance.

3 Record Audit and Reproducibility

3.1 Selecting one authoritative run

The main quantitative record is the recording-associated run 20260913_153855_682865. The earlier run 20260913_151835_077998 remains a separate comparison. Both completed five simulation cycles, but their coordinates and timestamps must not be merged. The main run's saved configuration has `autostart=false`, consistent with manual task initiation for filming; the earlier run uses `autostart=true`.

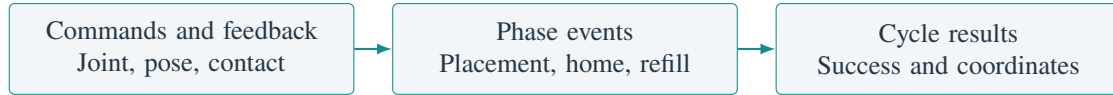


Figure 2: Complementary records used to check a completed simulation cycle.

Table 2: Evidence inventory for the main simulation run

Evidence	What it establishes	Count
results.json	Requested/completed cycles; placement and home outcomes.	5 / 5 cycles
events.jsonl	Phase transitions and verification order.	81 events
Placement events	Box accepted at B.	5
Home events	Return verified before success is recorded.	5
Refill request / ack	Replenishment between completed cycles.	4 / 4
trajectory.csv	Sampled robot/object feedback, including idle time.	4,475 rows

3.2 Checking order and elapsed time

The event log contains 60 phase entries, one run-start event and one completion event. No `SAFE_STOP` event appears. This supports normal completion, not successful testing of the stop branch. Four refill operations are expected: the fifth cycle ends the task instead of preparing a sixth object.

Initial-home entry to run completion spans approximately 318.02 s. The five saved cycle durations sum to 310.03 s; approximately 7.99 s lies outside those intervals, including refill-related transitions. The 329.13 s video contains recording time outside active execution. The trajectory also includes idle time, so its full span is not the task duration.

3.3 Keeping the evidence chain intact

I use result coordinates for numerical errors, events for ordering, and trajectory data for feedback history. Video supplies scene context but cannot independently establish millimetre-level placement. Analysis should retain the original B coordinate, compute at full precision, and round only displayed values. Each success must match placement and home checks, and each refill must follow home verification. An abandoned script's failure is development history, not a failure of the final five-cycle hardware run confirmed by the team.

4 Simulation Results and Error Interpretation

Table 3: Five-cycle main-run results, with home verified after every transfer

Cycle	Error (mm)	Heading (°)	Time (s)	Placed	Home
1	2.274	0.445	67.01	Yes	Yes
2	5.604	0.329	59.72	Yes	Yes
3	4.192	0.404	60.92	Yes	Yes
4	4.577	0.459	61.09	Yes	Yes
5	5.144	0.405	61.29	Yes	Yes

Source: Main-run results.json. Error is horizontal distance from B; heading is absolute home-heading error; time is the saved cycle wall duration.

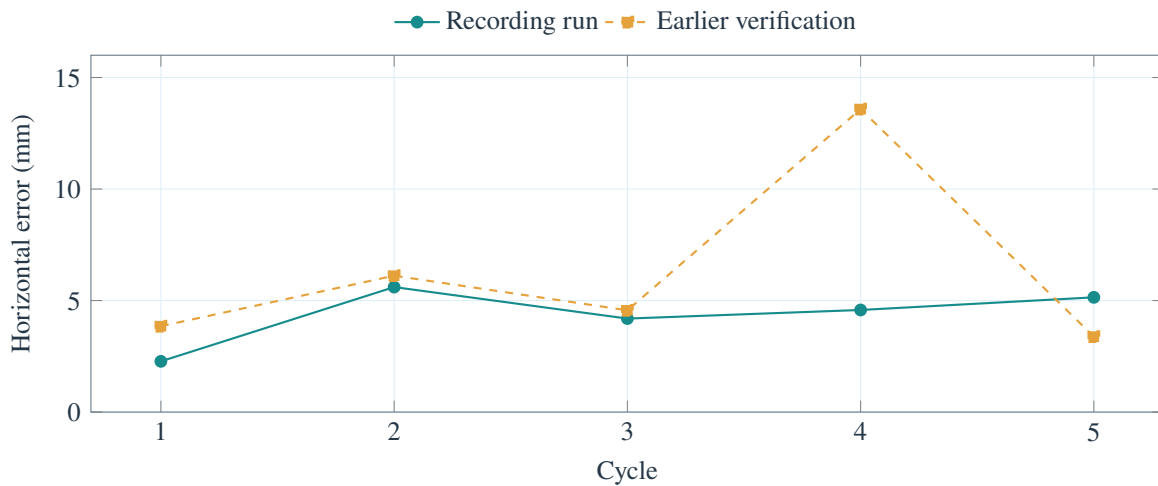


Figure 3: Two distinct five-cycle simulation runs. Neither series is hardware data.

Main-run result: 5/5 accepted placements and five verified returns home. Mean horizontal error is 4.358 mm; maximum error is 5.604 mm. Both are below the controller's configured 23 mm horizontal threshold.

The earlier run's mean was 6.299 mm, with a 13.585 mm fourth-cycle error. That larger value remains a successful placement under the same positional threshold and must not be omitted. The lower mean in the recording run does not prove a controlled algorithmic improvement: these are two small runs, not a randomized comparison with repeated conditions. The defensible conclusion is successful execution in both datasets with visible run-to-run variability.

The saved home arm error of about 4.21×10^{-14} rad reflects idealized numerical feedback, not physical positioning precision. I report this as a passed home condition rather than an advertised accuracy figure.

5 Hardware Outcome and Simulation Differences

5.1 Team-observed physical validation

The team confirmed that the hardware trials took place on 12 September 2026, with five successful transfers and no observed abnormalities using the final manual-confirmation workflow. Each lift was followed by a fresh human y confirmation before rotation; heading restoration and arm homing followed cycle 5. This is an observed 5/5 result, not an instrumented precision or timing measurement.

Table 4: Hardware outcomes confirmed by the team, not an automatic test log

Cycle	Direction	Outcome	Abnormality
1	A→B	Successful	None observed
2	B→A	Successful	None observed
3	A→B	Successful	None observed
4	B→A	Successful	None observed
5	A→B	Successful	None observed

5.2 Why the implementations must not be conflated

The archived hardware program uses the RoboMaster SDK, lifts from GRASP=(191, -51) to HOVER=(191, -1) in SDK millimetres, stops the chassis and waits up to 30 s for confirmation. The 50 mm difference is a commanded lift, not an independently measured object displacement. Odd cycles turn +32° and even cycles -32°. No confirmation means no transfer turn or automatic return; cleanup attempts to pause the gripper, stop the base and close the SDK connection. This is not a certified emergency-stop circuit.

Table 5: Differences retained in the analysis

Aspect	Final simulation	Final hardware
Control route	ROS 2, bridge and Gazebo controllers	Direct RoboMaster SDK
Direction	A→B each cycle	Alternating A/B
Grasp decision	Bilateral contact and object height	Human confirmation after lift
Return	Home verified every cycle	Initial home and final return after cycle 5
Replenishment	Four resets after home verification	No automatic cube refill
Evidence	Coordinates, events and trajectory	Team-observed successes

Do not transfer simulation precision to hardware. No measured physical placement errors, forces, durations or object mass were supplied. The model's 50 g cube mass is not a real weighing. Five successful transfers meet the observed success-count criterion for this sample, but do not establish long-run reliability or complete compliance with every course requirement.

6 Problems, Solutions and Individual Reflection

6.1 Version ambiguity and inconsistent claims

The development history includes object-animation experiments and an abandoned hardware checking script. Mixing these versions could produce contradictory claims about automatic object detection and manual confirmation. I anchor final behavior in the archived code and use conversation notes for development history. Simulation, final hardware outcomes and historical tests remain separate.

6.2 Success flags without sufficient context

A success flag is meaningful only when its conditions are understood. Here, placement checks horizontal distance, height and table contact; home verification follows separately. My solution is to cross-check coordinates against the target and join the result with the placement/home events, rather than treat either a closed gripper or a visually plausible video as sufficient evidence.

6.3 Offline tests versus system-level fault evidence

The fact-check record reports four passing offline tests on the final simulation code. One calls `Kinematics().solve(10,10,0.6)` and expects `ValueError` for an unreachable target. This establishes solver-level rejection, not the running system's stopping response. No complete final-version system-level fault-injection campaign is documented. Further tests should cover runtime target rejection, stale pose feedback, failed grasping and blocked refill, recording the expected error, observed state and stopping behavior separately.

6.4 A fair account of the adapted experiment

The recorded simulation and team-observed hardware both meet the success-count criterion. Direct SDK control and hardware end-only return nevertheless differ from the handout's shared-interface and parameter-only-switching objectives. The evidence also does not document each member's individual safety-operation check. I acknowledge these boundaries instead of equating 5/5 transfers with complete compliance.

6.5 Conclusion and next improvement

My contribution is an auditable account of the experiment, not a claim of personally programming or operating every subsystem. Experimental quality depends on traceable definitions as much as on a successful demonstration. A useful next step would attach a common trial identifier to video, telemetry and result files, then collect real placement measurements with a consistent reference. This would support a defensible simulation-to-hardware comparison without inventing precision.

Conclusion. The final records support five successful simulated cycles with per-cycle home verification and five team-confirmed physical transfers without observed abnormalities. Numerical precision is established only for the saved simulation sample; unrecorded tests remain unverified.

7 Shared GitHub Repository and Commit Evidence

The complete group archive is stored in the [shared public GitHub repository](#). The repository address is github.com/ccx8kg5qpd-debug/robomaster-ep-fixed-point-grasping. It contains the simulation and hardware programs, model resources, saved results, the complete simulation recording and all five individual report sources.

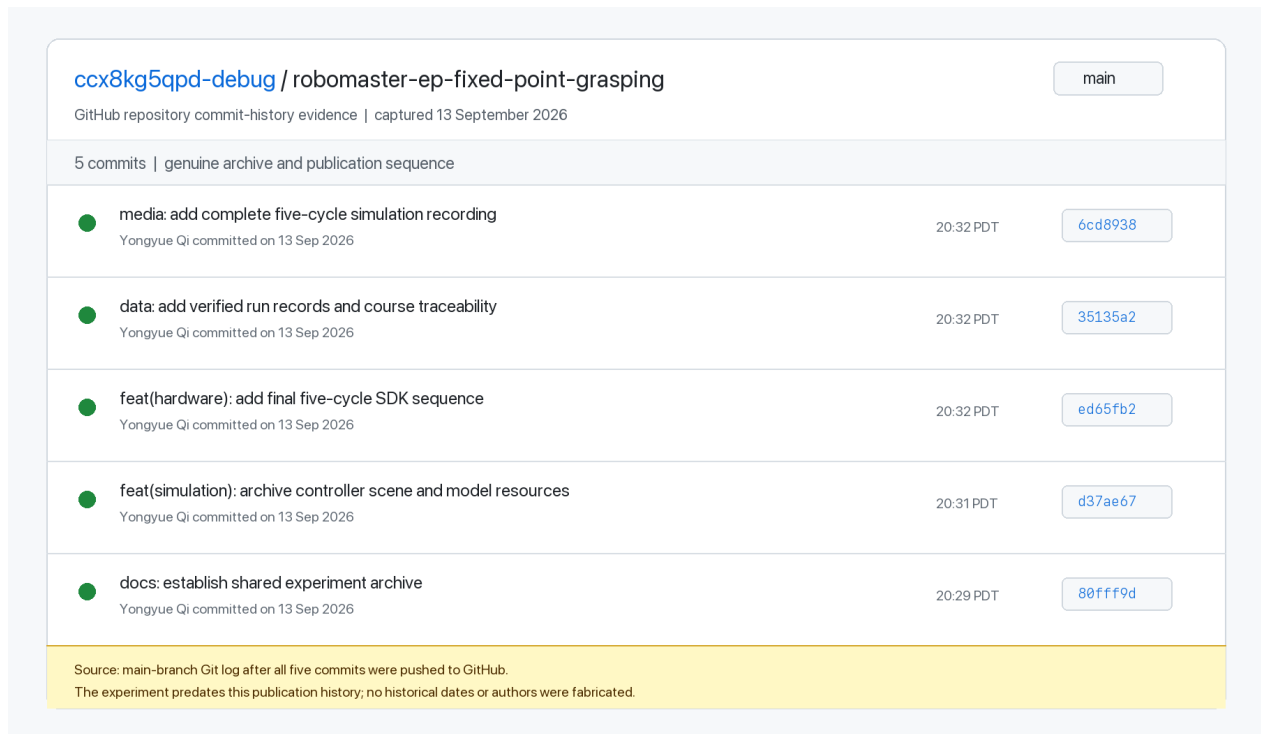


Figure 4: Commit-history evidence captured after the first five project-material commits had been pushed to the shared GitHub repository.

Traceability note. The experiment was completed before the GitHub repository was assembled. These genuine commits document the later review, organization and publication sequence; they are not presented as a fabricated contemporaneous development history. The image records the five commits visible at capture time, before this revised report was added.

8 Sources, Reproduction and Evidence Limits

8.1 Sources and traceability

[1] Current course handout, *Robotics Integration Group Project I: Fixed-Point Robotic-Arm Grasping, Simulation Development and Hardware Validation*, S16–S20 / E5, and the individual-report assignment requirements. The course field is translated from this experiment’s handout.

[2] Shared team repository, [robomaster-ep-fixed-point-grasping](#), assembled 13 September 2026: simulation source/configuration, `real_v2_five_cycles_manual.py`, saved results, event log and trajectory. Main run: 20260913_153855_682865; comparison run: 20260913_151835_077998.

[3] Team conversation fact-check materials dated 13 September 2026 for development and offline-test history; subsequent team confirmation for the successful final five-cycle hardware run.

[4] Community model resources: [jeguzzi/robomaster_ros](#), archived commit c05a39d7f0fa. This dependency repository is not the team’s experiment repository.

[5] Author’s QYY personal layout, `Elegant_Project_Report_Template.tex`. Earlier QYY work was consulted for visual design only. No previous experiment’s body, images, metrics or conclusions were reused.

8.2 Reproducing the numerical summary

Unmodified copies of both result files, the main event log and its trajectory are included in `supporting_data/`. The packaged `check_numbers.py` recomputes errors, cycle totals and event counts without starting a simulator or connecting to a robot. The analysis below illustrates the same calculation; it is report-analysis code, not the robot controller.

Listing 1: Horizontal-error calculation from the saved main-run record

```
import json, math, statistics
from pathlib import Path
run = "20260913_153855_682865"
path = Path("supporting_data") / run / "results.json"
data = json.loads(path.read_text())
bx, by, _ = data["configuration"]["b_world_m"]
errors_mm = []
for cycle in data["cycles"]:
    x, y, _ = cycle["placement_world_m"]
    errors_mm.append(1000 * math.hypot(x - bx, y - by))
print(statistics.mean(errors_mm), max(errors_mm))
```

8.3 Record availability and experimental limitations

- **Repository:** the group uses the shared public repository linked in Section 7. Its publication commits were made after the experiment and are described as archive history, not experimental-development history. No previous-project repository is substituted.
- **Dates:** the team-confirmed hardware trial date is 12 September 2026. The cover date, 13 September 2026, is the report-preparation date.
- **Evidence limits:** no additional physical placement, timing, force or mass measurements, individual safety-check records or system-level fault-test records were supplied. These properties are not inferred from the team-observed 5/5 outcome. The offline-test evidence described in Section 6 remains distinct from unrecorded system-level tests.